SUVIN POOJARY

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PROFESSIONAL SUMMARY

Final-year Artificial Intelligence & Machine Learning undergraduate with hands-on experience building AI applications across healthcare, large language models, and data-driven systems. Experienced in retrieval-augmented generation (RAG), vector databases, SQL-based data retrieval, and local LLM deployment. Currently developing a multimodal deep learning framework integrating CT imaging and clinical data for kidney tumor risk assessment, with a focus on reliable AI and applied machine learning.

RESEARCH INTERESTS

AI for Healthcare · Multimodal Learning · LLMs, NLP & RAG · Reliable & Explainable AI

EDUCATION

NMAM Institute of Technology (Nitte), Udupi *August 2023 – Present*

Bachelor of Technology in Artificial Intelligence and Machine Learning

- CGPA: 8.46 / 10.0
- Coursework: Neural Networks, Deep Learning, MLOps, Cloud Computing, DSA

ACADEMIC RESEARCH

MetaFusion-Driven Multimodal Learning for Kidney Tumor Risk Assessment

Final Year Research Project *June 2026 – Present*

- Developing a multimodal framework that fuses abdominal CT imaging with 12 pre-operative clinical features from the KNIGHT Challenge dataset to stratify kidney tumor risk.
- Adapted the MetaFusion architecture with a ResNet-18 image backbone and clinical feature encoder, supporting both five-way risk classification and binary high/very-high-risk detection.
- Built the CT preprocessing pipeline: tumor-mask-guided selection of the axial slice with the largest tumor cross-section, followed by soft-tissue CT windowing, turning 3D volumes into model-ready 2D input.
- Benchmarking against image-only, metadata-only and fusion baselines (MetaBlock, MetaNet) with 5-fold cross-validation, reporting balanced accuracy, macro-F1 and AUC-ROC to test whether fusion improves on either modality alone.

EXPERIENCE

Student Intern — Manipal School of Information Sciences (MSIS), MAHE *Jun 2026 – Aug 2026*

- Restored and redeployed two hospital chatbots fully on-premises using FastAPI, Docker, Qdrant and Ollama-hosted local LLMs, removing dependence on external AI services so patient data stays within the hospital network.
- Reworked retrieval and database-query logic so answers draw on the right records, and added chat caching so repeat questions are served faster, along with suggested questions, logging and AI summarization.
- Built conversational report generation and Patient Search integration that return structured patient summaries on request.
- Implemented security guardrails that block unsafe queries before they reach sensitive patient data; verified every workflow through end-to-end testing and backend log analysis.

Summer Research Intern — NMAM Institute of Technology *May 2024*

- Researched applications of machine learning in medical and pharmaceutical sciences under the guidance of a lecturer, and presented the findings.

PROJECTS

PharmaGuard — Evidence-Driven AI Pharmaceutical Safety System *In Development*

- Building a drug-safety decision-support platform that ingests adverse drug reaction (ADR) reports and combines LLM/NLP with rule-based reasoning to produce explainable, evidence-backed alerts, routed through a pharmacist approval workflow with a full audit trail.
- Designed a traceable six-stage LLM pipeline (de-identification, extraction, classification, RxNorm medication normalization, RAG evidence retrieval, summarization) so any wrong alert can be traced to the stage that caused it.
- Implemented a PII de-identification module and a RAG layer (Sentence-Transformers + Qdrant) that surfaces similar past cases and drug-label evidence with similarity scores and citations, so pharmacists can verify the evidence behind each alert.
- Architected as three independently deployable containerized services: React client, Express API and FastAPI ML microservice.

HoloGen 3D-Reconstructor

- Developed a model to convert 2D images into 3D scenes using PyTorch, OpenCV, and PyRender.
- Focused on monocular depth estimation for accurate 3D reconstruction and visualization.
- Implemented image preprocessing and depth estimation workflows to generate depth information from single RGB images.
- Generated and visualized reconstructed 3D scenes using estimated depth maps, enabling spatial representation of objects from 2D inputs.

ApexPitwall — ML-Powered F1 Pit-Stop Strategy Simulator

- Built a five-stage pipeline (lap-time prediction, tyre degradation, strategy simulation, counterfactuals, Monte Carlo) on FastF1 data from five 2022 Grands Prix, with leakage-safe features.
- Trained Ridge and LightGBM lap-time models; LightGBM reached 0.46 s MAE (R^2 0.78) in 5-fold CV on 1,060 Catalunya laps, 22% better than a rolling-pace baseline.
- Simulated 118 one- and two-stop strategies using fuel-adjusted tyre degradation and a measured 22.59 s pit loss; Soft-Medium-Soft beat one-stop options by 32.8–44.3 s. Added Monte Carlo uncertainty and a Streamlit dashboard.

WaveHead

- Built a team-based music player app with MySQL database integration for managing users and playlists.
- Designed an intuitive UI and ensured seamless audio playback.

SKILLS

- **Programming:** Python, C++, SQL, JavaScript
- **AI & Machine Learning:** Deep Learning, Computer Vision, NLP, LLMs, RAG, Multimodal Fusion, Monocular Depth Estimation, Cross-Validation & Benchmarking
- **LLM & Retrieval Stack:** Ollama (local LLMs), Qdrant, Sentence-Transformers, Prompt Pipelines, Guardrails, PII De-identification
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, LightGBM, OpenCV, U-Net, PyRender, Pandas, NumPy
- **Backend & DevOps:** FastAPI, Node.js/Express, React, Streamlit, Docker, Git/GitHub
- **Tools:** Jupyter, Google Colab, VS Code
- **Languages:** Fluency in English, Hindi, Kannada

CERTIFICATIONS

- Google Cloud Computing Foundations — NPTEL (Aug – Oct 2024)